

The Cost of One Operator: Structural Expansion, Compression, and Pilot Learning in Generated Symbolic Expressions

Anonymous ACL submission

Abstract

One binary operator can, in principle, represent generated elementary symbolic expressions: [Odrzywołek \(2026\)](#) constructs every elementary function from 1 and $\text{eml}(x, y) = \exp(x) - \ln(y)$. We call the resulting study, benchmark, and implementation GEML (Graph Learning over a Single-Operator Representation of Mathematics). We profile a fixed synthetic corpus of 250,000 deterministic expressions compiled by the pinned official compiler. Structural node counts are exact under that compiler and canonicalization; semantic agreement is bounded to an audit with 6/3,261 mismatches and one deterministic numeric hang. Median tree expansion is $\alpha = 40.66$, above every preregistered family threshold (1.29–1.50). Lossless DAG sharing compresses pure-EML trees by $39.4\times$ but leaves them $8.3\times$ the uncompressed AST tree and $10.5\times$ the AST DAG on average. Verified rewriting improves 23.9–27.6% of costed rows by 2.4–2.8% on 60.7% coverage, conditional on the covered families. Equal-budget transparent baselines beat both learned components. Two of three equivalence runs obtain validation BCE below a descriptive class-prior baseline, but comparative representation quality remains untested. A goal-conditioned value-head diagnostic finds near-zero IID rank correlation and a constant out-of-distribution predictor. Anonymized code and audit artifacts accompany the submission.

1 Introduction

Can one operator be useful for learning on generated elementary symbolic expressions? Machine encodings trade vocabulary for structure, affecting formula retrieval ([Lample and Charton, 2020](#); [Mansouri et al., 2019](#)) and graph inputs for neuro-symbolic search ([Allamanis et al., 2018](#); [Irving et al., 2016](#)). Pure EML is the maximally regular endpoint: a graph network need not learn *what* an operator is, only *where* things connect. The objection is size (even $\sin(x)$ expands to hundreds of

nodes); DAGs, macros, and rewriting might recover that cost. GEML answers the structural question with exact counts and reports bounded learning diagnostics. The equivalence pilot finds a small validation signal in two seeds, while the rank/OOD diagnostics locate what is still missing: usable guidance.

This is a structural feasibility study with preregistered thresholds and complete denominators. We measure a fixed synthetic corpus of 250,000 expressions but make no scaling, leaderboard, or frontier-LLM claim; each number below is bound to a committed result summary.

Our novelty is a controlled recovery ladder—source AST, EML tree, exact-subterm DAG, verified rewriting, and transparent or learned vocabulary compression—rather than a generic compression-ratio report.

Contributions.

- An exact structural cost profile on a fixed synthetic corpus of 250,000 expressions: median expansion $\alpha = 40.66$, per-family threshold comparisons, and content-hashed controls in which transparent baselines remain unbeaten (§6).
- A bounded pure-EML-DAG equivalence pilot: two of three runs obtain validation BCE below a descriptive class-prior baseline, indicating a small nontrivial signal; comparative representation quality remains untested (§6.5).
- A transferable methodological result: preregistered rank and OOD diagnostics expose a guidance failure hidden by proxy error ([Burges et al., 2005](#)) (§6.5).

Resources. The GEML implementation is available in the [anonymized code repository](#). Corpus and result archives, manifests, rule sets, verification records, and reproduction instructions are available

in the [anonymized artifact repository](#). The main paper states all claims and denominators; the repositories support audit and reproduction.

2 Related work

Encoding generated symbolic expressions for learning. Mathematical language uses prefix/infix tokens (Lample and Charton, 2020), formula embeddings and retrieval (Mansouri et al., 2019), operator trees, and shared term graphs (Al-lamanis et al., 2018). These expose a vocabulary–structure tradeoff: universal or low-vocabulary encodings push complexity into structure, with the single-operator construction (Odrzywołek, 2026) as an extreme case. Subsequent EML work studies architecture-dependent recoverability in differentiable symbolic regression (Gupta and LaGrow, 2026) and universal approximation by parameterized EML trees (Germany et al., 2026). Those works concern direct fitting or expressivity; GEML measures compiled structural cost and recovery through sharing, verified rewriting, and vocabulary induction.

Sharing, rewriting, and guidance. Hash-consing and term-graph reduction (Ershov, 1958; Filliâtre and Conchon, 2006; Barendregt et al., 1987), equality saturation (Tate et al., 2009; Willsey et al., 2021), and MDL vocabulary induction (Rissanen, 1978) are recovery hypotheses. Neural premise selection, proof search, and symbolic regression (Irving et al., 2016; Polu and Sutskever, 2020; Lample et al., 2022) consume learned guidance; learning-to-rank separates regression and ranking (Borges et al., 2005).

3 Representations, immutable data, and hypotheses

3.1 Representations

Every representation we compare derives from the same source expression, so differences between them are differences of encoding rather than of content. The structural study (§6) uses four: the ordered source AST as a tree and as a hash-consed DAG (Ershov, 1958; Filliâtre and Conchon, 2006), and the official-v4 pure-EML term (produced by the pinned version-4 reference compiler) as a tree and as a hash-consed DAG. Writing α for the ratio of exact pure-EML tree nodes to source AST nodes, the tree pair fixes α and the DAG pair determines what maximal exact-sharing under the

chosen canonical signature recovers from it.

The frozen comparison schema specifies six arms: AST DAG, pure-EML DAG, macro-derived frequent-motif DAG, train-only motif-AST, prefix tokens, and a floor of 19 surface counts. The authenticated GPU evidence contains pure-EML-DAG source cells only, so the matched representation comparison was not completed; the feature list is frozen in code and asserted by tests.

Macro and motif graphs are lossless DAGs over admitted source constructions, not pure-EML graphs: each macro node has `is_pure_eml = false`, while its pure-EML expansion remains separate metadata. Repeated references are retained, and structural identity is hash equality on a canonical node signature; semantic equivalence is a separate, verifier-established relation.

3.2 Fixed data foundation

The v1 corpus has 250,000 expressions, unique under canonical (`domain_mode`, `sympy_srepr`) identity (Meurer et al., 2017), with splits 175,000/25,000/25,000/25,000 (train/validation/test-IID/OOD) and exact family quotas: 70,000 algebraic core, 40,000 each exp/log, powers/division/rationals, and trig/hyperbolic, 35,000 mixed elementary, and 25,000 OOD stress.¹ Generation is deterministic and QA-gated; manifests carry shard checksums, config hash, seed, and commit.

The OOD family is exactly the held-out test_ood split. It combines stress over size, depth, variable count, and composition, not a depth-controlled holdout; all claims are restricted to this v1 corpus.

3.3 Predeclared outcome families and hypotheses

Outcome families were preregistered and reported separately: structural (tree/DAG size and depth, pure-EML α , macro size, and dictionary-inclusive MDL (Rissanen, 1978)), predictive (the six-arm equivalence grid), rewrite/proof, symbolic regression, and grammar-v2 conformance; no pooled score is used. For the preregistered counting model, a full binary source tree with x internal nodes has $C_x K^x L^{x+1}$ labeled forms, whereas a single-operator tree with αx internal nodes has $C_{\alpha x} L^{\alpha x+1}$. Since $\ln C_x/x \rightarrow \ln 4$, their asymptotic description lengths break even at $\alpha_{\text{tree}} =$

¹[anonymized artifact repository](#): Corpus QA report and Goal 2 summary, “Corpus and denominators”.

$1 + \ln K / \ln(4L)$. H1 uses the strict inequality below this value (threshold range 1.2914–1.5000). Here K is the declared source operator-label count and L is the declared bounded leaf-label count; the full leaf vocabulary contains six variables, 19 integers from -9 through 9 , and 92 distinct reduced nonintegral rationals. Because empirical α counts total nodes, H1 is a preregistered asymptotic coding scenario, not a finite-tree or semantic theorem. H2 tests whether exact sharing reaches AST parity, and H3 tests verified rewriting and learned motifs. Structural statistics use all 250,000 rows; semantic auditing is a separate bounded sample (§9).

4 Compact models and fair compute

4.1 A compact GINE-style graph encoder

The reported equivalence cells use a frozen, project-specific GINE-style encoder (Xu et al., 2019; Hu et al., 2020): three message-passing layers, hidden width 64, residual updates, slot/role embeddings, LayerNorm, dropout 0.1, a two-layer feed-forward block, a virtual node, and sum pooling. The node-label table has 4,096 entries. A shared Siamese encoder combines endpoint embeddings by sum, absolute difference, and elementwise product, then applies a two-layer binary-classification head. The complete equivalence model has 456,452 trainable parameters. The matched six-arm grid was not completed, so no cross-representation quality claim is made.

4.2 Task composition and controls

The symmetric endpoint composition is invariant to swapping a, b . The transparent floor uses 19 predeclared counts (operators, primitives, variables, size, and depth). Vocabulary embeddings count toward a matched parameter budget because label vocabularies differ by channel; only the pure-EML-DAG graph cells were completed in this release.

4.3 Compute matching and training

Pure-EML graphs are much larger than AST or macro graphs, so equal parameters are not equal compute. All three equivalence cells use AdamW (learning rate 10^{-3} , weight decay 0.01, gradient clipping at 1.0), at most 12 epochs or 3,000 optimizer steps, and validation-loss early stopping with patience 3. Dynamic batches are capped at 16,384 nodes, 65,536 directed message edges, and 128 pairs. The frozen H100 profile requests bfloat16 autotcast and enables TF32; the run ledger records ef-

fective precision, time, memory, GPU-hours, graph sizes, and deterministic settings per cell.

5 Bounded tasks and verification

We evaluate representation quality through bounded, verifier-gated tasks rather than an aggregate score. Each task has frozen identifiers, declared budgets, retained failures, and explicit denominators; unavailable cells are missing, not zero.

5.1 Equivalence

The production equivalence specification fixes 50,000/5,000/5,000/5,000 accepted records (train/validation/test-IID/test-OOD). Each record is a replayable rewrite trajectory with rule, direction, occurrence path, bindings, assumptions, source/successor signatures, and verifier evidence. Source-, e-class-, and trace-relative grouping prevents leakage; rejected, invalid, unsupported, and timeout candidates remain typed ledger rows. All four graph channels and two controls use the same endpoints, and IID/OOD/depth/family views never alter base split totals.

The byte-identical-twice 10,000-row pilot used the real rule engine and replay verifier: 9,039 records in 34 CPU minutes (6,867 verified positives, 2,172 exact-rational negatives), plus 2,824 rejections and 3,133 typed failures. It is complete but below target, so no resampling is hidden. Accepted train/validation/test-IID/test-OOD counts are 6,281/893/864/1,001; rejected 1,786/263/276/499; failures 2,432/343/358/0; shortfalls 43,719/4,107/4,136/3,999. The OOD split has one accepted negative because same-family size-matched neighbors are scarce; the imbalance is retained.

5.2 Further bounded tracks (future work)

The remaining tracks contribute no headline numbers here. Rewrite-step prediction is partial (13/18 logical cells, with five invalid cells retained); proof (256 problems), simplification (1,000 identifiers), and symbolic regression (256 synthetic tasks plus a 32-task Feynman subset (Udrescu and Tegmark, 2020)) require verifier-confirmed recovery and retain failure costs. The Pareto synthesis was not run.

6 Results at fixed scale

All structural values come from recorded CPU summaries; the equivalence and value-head verdicts bind to the committed, digest-verified GPU run ledger. None is estimated. We organize by outcome rather than by a single leaderboard and use the figures below for the headline comparisons.

6.1 Raw expansion on the fixed corpus

All 250,000 expressions compiled successfully under the pinned official-v4 compiler. The tree expansion factor α spans 1.50 to 6,481,679 with median 40.66 (mean 952.14; full distribution in the artifact distribution).² The mean/median gap is a heavy right tail dominated by the trig/hyperbolic family (median 270.46), not a typical 952-fold expansion. Figure 1 shows every family against its preregistered threshold: all six miss by one to two orders of magnitude, so the strict pass count is 0/250,000. The full distribution is recorded in-line for reproducibility: p10 21.0338, p25 26.3800, median 40.6602, p75 75.2550, p90 385.0723, p95 1,187.8240, and p99 10,448.5978; the maximum is 6,481,679.3077 and the standard deviation is 23,155.5594. *Takeaway.* Universal regularity shifts burden from vocabulary into structural length; this is a representation-level constraint, not compilation failure.

6.2 Lossless sharing and the remaining gap

Lossless hash-consing with maximal exact-sharing under the chosen canonical signature compresses expanded EML trees by a mean factor of 39.3750, yet the shared EML DAG remains larger than the *uncompressed* AST tree for every one of the 250,000 expressions; the best remaining ratio is exactly 8/7.³ The most compressible expression shrinks by 28,087,277/180 and still ends 540/13 times larger than its AST (mean EML-DAG/AST-tree 8.3344, EML-DAG/AST-DAG 10.4750). Compression does not imply structural competitiveness. *Takeaway.* Hash-consing recovers syntax, but a first-order DAG/AST gap remains for later compression stages to justify.

²anonymized artifact repository: Goal 2 summary, “Result” and “Expansion distribution”.

³anonymized artifact repository: Goal 3 summary, “Structural verdict”.

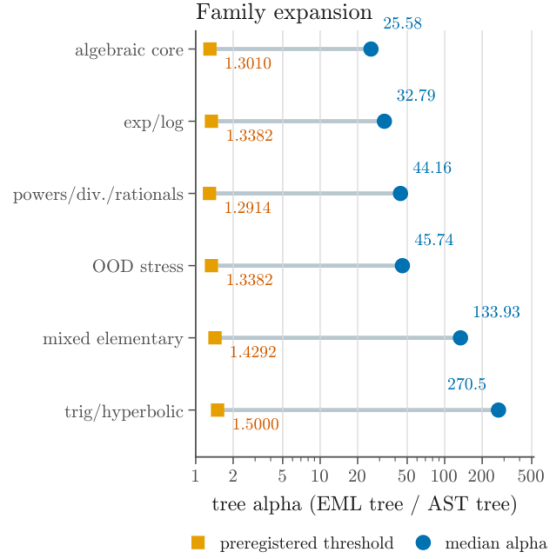


Figure 1: Raw tree expansion measured over all 250,000 expressions. Each row compares the family median EML-tree/AST-tree expansion α (blue circle; larger is worse) with its preregistered strict counting threshold (orange square; thresholds 1.2914–1.5000). Families are shown in the order algebraic core (70,000), exp/log (40,000), powers/division/rationals (40,000), OOD stress (25,000), mixed elementary (35,000), and trig/hyperbolic (40,000). Every one of the 250,000 rows has zero strict passes; the OOD family is the fixed held-out stress split, not a depth-only holdout.

6.3 Verified rewriting adds conditional savings

On a frozen, stratified 30,000-expression selection, we run bounded equality saturation (Tate et al., 2009; Willsey et al., 2021) with complete application provenance and independent candidate verification. It improves 4,349/18,210 costed rows (23.882%) under branch-insensitive real rewrites (safe_real) and 5,026/18,210 (27.600%) with guarded positive-real rules (positive_real_formal), at mean relative savings of 2.361% and 2.849% over costed rows and a largest single reduction of 302 exact EML-DAG nodes.⁴ Figure 2 summarizes the recovery ladder. Coverage is the binding caveat: 11,566 rows per mode (38.553%) contain operators outside the closed rewrite vocabulary (including all trig/hyperbolic and mixed-elementary rows), so these results hold on 60.700% cost coverage and are conditional on it. The 2.4–2.8% savings therefore apply only to covered, least-pathological

⁴anonymized artifact repository: Goal 4 summary, “Production result”.

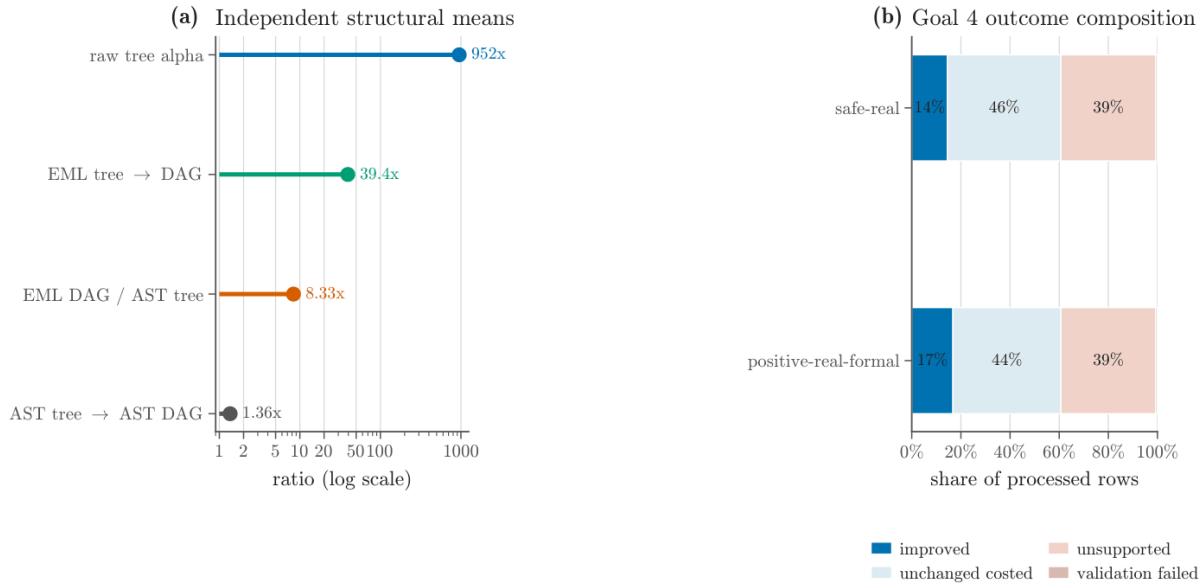


Figure 2: Structural recovery and verified rewriting. **Left:** independent structural mean ratios (lower is closer to the uncompressed AST): raw EML-tree/AST 952 $\times$, EML tree-to-DAG 39.4 $\times$, EML DAG/AST tree 8.33 $\times$, and AST tree-to-DAG 1.36 $\times$. **Right:** rewrite-study shares of all 30,000 rows; only 18,210 (60.7%) are costed, so improved/costed rates are conditional (23.882% and 27.600%). Unsupported/failing rows remain; zero timeouts.

supported families; trig/hyperbolic, the largest-expansion family, and mixed elementary are excluded. Each mode retains 224 validation failures (0.747%) and zero timeouts; the source form is always a costed safety anchor, so improvement rates cannot be inflated by dropped rows. *Takeaway.* Verified rewriting recovers bounded savings without erasing structural cost.

6.4 Learned components do not yet beat transparent baselines

Both learned-component comparisons were published in frozen null-before-positive order.⁵ Figure 3 gives the nulls. A learned motif vocabulary did not beat the equal-budget frequent-motif vocabulary on test_iid dictionary-inclusive MDL: 324,485,346 vs. 317,678,264 bits over 25,000 expressions (it did beat the uncompressed macro baseline and median fixed-random vocabularies). A neural cost ranker also did not beat structural heuristics: 8,349/18,050 exact-best rankings vs. 8,796/18,050 for the estimated-EML-tree-cost heuristic (and 8,661/18,050 for the AST-DAG-cost heuristic). Counting is conservative: of the 18,050 groups, 10,752 were evaluable, and every unevaluable group scores as a miss for every ranker. Transparent structural baselines remain unbeaten at equal

⁵anonymized artifact repository: Goal 5 summary, “Learned and neural comparisons”.

budget. The OOD deterministic random result is $1,078/2,034 = 53.0\%$ exact-best among evaluable groups. Exact-best is one signature selected under deterministic rewrite-study tie-breaking; candidate-count and cost-tie histograms were not published, so this value is not interpreted as calibrated random chance.

H3 is therefore not supported at this scale: verified rewriting recovers 2.4–2.8% mean savings on 60.7% cost coverage, and neither the learned motif vocabulary nor the neural ranker beats its equal-budget transparent counterpart. The residual gap left by H1 and H2 remains open. *Takeaway.* Transparent structural signals remain the useful equal-budget reference; no learning advantage is demonstrated here.

6.5 Learning tracks: bounded pilot and nulls

Every number in this subsection binds to the committed, digest-verified GPU run ledger. Two tracks are measured (equivalence and the value head); rewrite-step prediction is partial, while symbolic regression and Pareto synthesis were not run (§5.2). The measured tracks yield one bounded viability result and one methodological null.

The equivalence track is a bounded viability pilot (Figure 4) on pure-EML-DAG source cells (three seeds, no AST arm): two seeds reach BCE 0.5267/0.5387, while the third diverges to 2.3914.

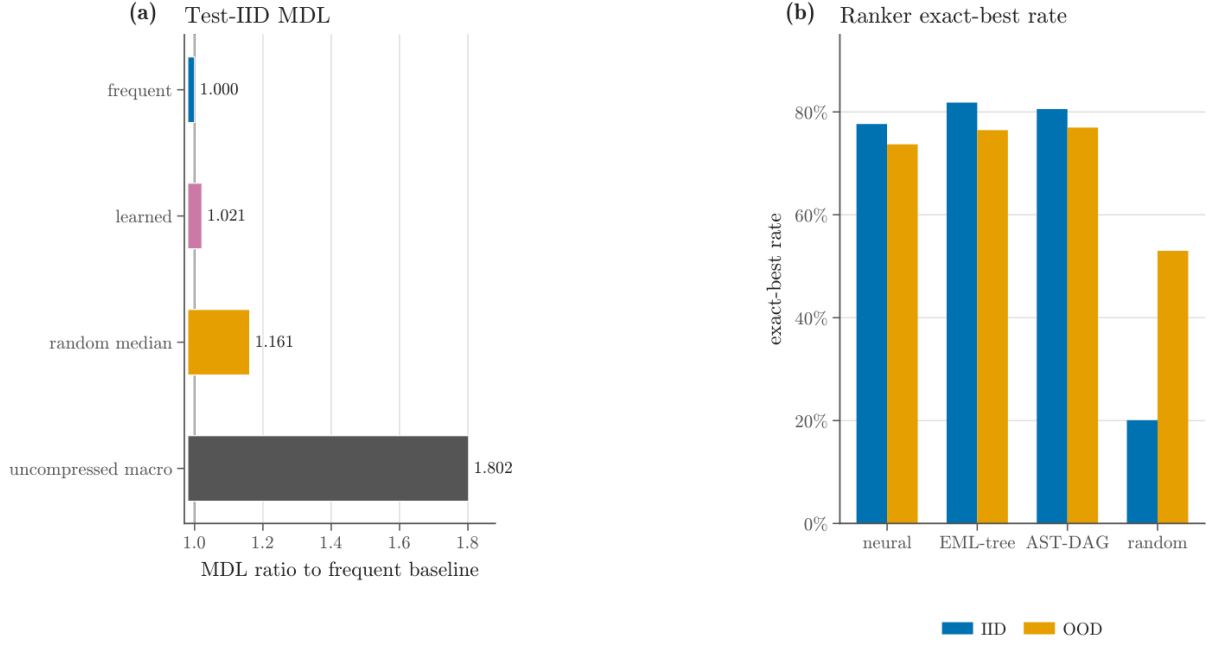


Figure 3: Learned-component null comparisons. **Left:** dictionary- inclusive test-IID MDL ratios (lower is better) for 25,000 expressions: learned 1.021, frequent 1.000, random 1.161, uncompressed 1.802. **Right:** exact-best rates (higher is better) among evaluable IID/OOD groups (10,752/18,050 and 2,034/2,214). The preregistered prose uses all 18,050 groups and counts unevaluable rows as misses; the neural ranker loses to the transparent EML-cost heuristic on that denominator.

Accepted pairs are imbalanced (validation 657/893 positives), so a constant class-prior (majority-rate probability) predictor gives ≈ 0.577 BCE; this floor is computed from that validation composition and is descriptive, not independent. Two of three pilot runs obtain validation BCE below a class-prior baseline, indicating a small nontrivial signal; comparative representation quality remains untested. The pure-EML-DAG means are 8.3344 versus AST tree and 10.4750 versus AST DAG; raw 40.66 is structural only. The six-arm grid was not run. All three cells use the frozen 456,452-parameter width-64, three-layer, virtual-node configuration in §4; seed-3 divergence has no resolved cause.

The value-head track shows what rank/OOD diagnostics catch that error alone does not (Figure 5). It predicts *remaining witness steps*, the verified suffix length for an ordered (current, goal) pair: a witnessed upper bound, not shortest-path/admissible distance. Alternate witnesses are not proved to share labels, so interpretation is path-conditioned. Under a reduced budget (1,500 steps, three seeds, 452,820 parameters, exp_log held out), validation MAE is 0.830–0.847 and test-IID MAE 1.87–1.92. No predefined acceptance threshold existed, so low MAE is descriptive. Test-IID Spearman is -0.014 , 0.002 , and -0.009 ; the OOD predictor is constant

(undefined Spearman), with constant MAE 1.86–1.91 matching the trained head’s test-IID error. Error cannot separate the head from a constant; the run is therefore a null, not a proof-search heuristic.

7 Discussion

The main transferable GEML result is a representation-level tradeoff: a universal operator buys a tiny vocabulary and uniform local semantics but shifts cost into graph length; hash-consing, verified rewriting, and dictionary-inclusive pricing recover only part of that burden. This ladder separates encoding regularity from the structural budget available to downstream models of generated expressions.

Claim–evidence scope. Expansion, sharing, rewriting, and vocabulary claims bind to corpus, paired-DAG, frozen-selection, and MDL ledgers; the methodological claim binds only to the value-head rank/OOD diagnostic. These boundaries are not pooled into one score.

The value-head diagnosis is complementary. Neural proof and symbolic-regression search select value/cost heads by proxy (Polu and Sutskever, 2020; Lample et al., 2022); ours has unremarkable error but near-zero ranking and a constant OOD

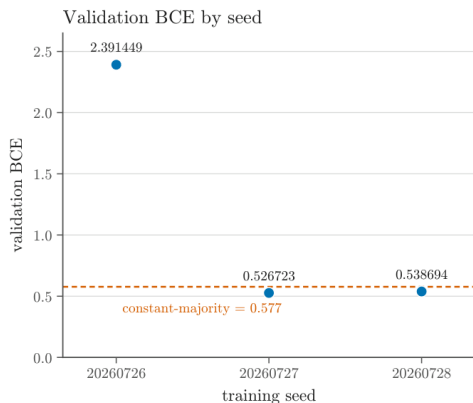


Figure 4: Equivalence viability pilot: validation binary-cross-entropy (BCE; lower is better) for the pure-EML-DAG arm. The imbalanced pilot floor is 0.577; seeds score 0.5267, 0.5387, and 2.3914. The frozen schema records BCE only—not accuracy or AUC—so this is viability, not established classification quality.

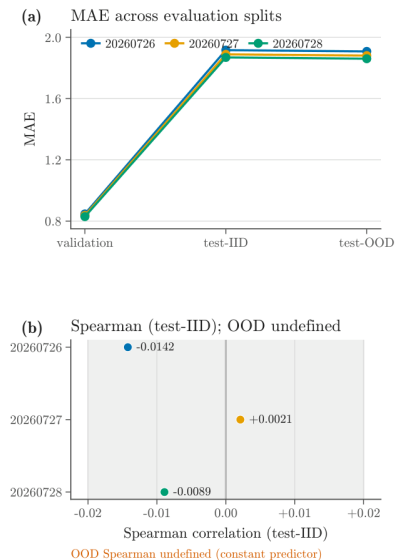


Figure 5: Value-head diagnostics across three seeds. **Top:** MAE (lower is better) for remaining-witness steps (maximum four): validation 0.830–0.847, test-IID 1.87–1.92. **Bottom:** test-IID Spearman is -0.014 , 0.002 , and -0.009 (near zero); OOD is constant and undefined. Constant OOD MAE (1.86–1.91) matches test-IID error, a guidance null.

predictor matching test-IID error. No predefined acceptance threshold existed: rank/OOD probes expose a heuristic that low error alone could make look favorable, leaving usable guidance an open frontier.

8 Conclusion

On the fixed synthetic corpus of 250,000 expressions, GEML delivers an exact structural cost profile under the pinned compiler, a bounded semantic audit, and a frozen learning apparatus. Median expansion is $40.66\times$; lossless sharing recovers $39.4\times$, and verified rewriting adds 2.4–2.8% savings on 60.7% coverage. Two of three equivalence runs obtain validation BCE below a descriptive class-prior baseline, but comparative representation quality remains untested. Preregistered rank and OOD diagnostics leave usable guidance as the remaining frontier.

9 Limitations

All measures use one fixed synthetic corpus of 250,000 generated elementary symbolic expressions; no scaling law is fit and three-seed effects are descriptive. The source AST is binary: variadic SymPy Add/Mul nodes are deterministi-

cally left-folded, so ratios use a binarized control, not native n -ary ASTs. Numeric verification is sample-bounded: the structural audit (273 rows) plus widened pilot (2,988) retains 6/3,261 mismatches ($\approx 0.2\%$) and one deterministic numeric hang. Three final-corpus mismatches are consistent with extreme intermediates, cancellation, and backend range/precision; three widened-pilot mismatches remain unclassified, so no corpus-wide semantic error rate is inferred.⁶ Rewriting savings are conditional on 60.7% coverage and exclude trig/hyperbolic expressions. The matched AST/prefix/motif/19-feature grid was not completed, so comparative quality is untested. Equivalence reports validation BCE only and one of three seeds diverges; the value head uses one reduced budget and path-conditioned targets. The grammar-v2 conformance check fails on eight `asin/acos` endpoint cells that are nonfinite at 100-digit precision under 10^{-60} tolerances, byte-identical across three executions, with no integrity, coverage, or v1-drift failures.⁷ The anonymized code and ar-

⁶anonymized artifact repository: Goal 2 summary, “Semantic audit and retained issues”.

⁷anonymized artifact repository: Goal 10 summary, “Published verdict”.

tifact repositories provide downloadable instructions, manifests, rule sets, verification records, and reproduction commands; SHA-256 hashes authenticate downloaded bytes. These limits leave two questions: which sharing or vocabulary choices preserve semantic signal at lower graph cost, and which diagnostics turn it into search guidance.

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